SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 1

Teaching AI to Check Itself



The Reflection Pattern: an AI drafts a forensic report, critiques it, then revises it, so every claim stays inside what the evidence supports.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

Eight steps, one careful habit

1

What is an agent

A model that follows a loop, not a one-shot answer.

2

Before AI

How a report was checked by a human reviewer.

3

The problem

A single AI answer sounds sure, and overclaims.

4

The discipline

Cite every claim. Separate fact from guess.

5

The gap

Who reviews the AI when no human is in time?

6

Reflection

The AI critiques and revises its own draft.

7

The case

A phone, a leaked file, and six artifacts.

8

Your lab

Run the loop yourself, free, in the browser.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

FROM A CHATBOT TO AN INVESTIGATOR

One answer, or a working loop

A plain model call

You ask once. The model answers once. If the answer overreaches, nothing catches it.

1. You write a prompt
2. The model replies
3. It stops

TODAY

An agent

You give the model a goal and a role. It works in a loop: produce, check, and improve before it hands anything back.

1. Produce a draft
2. Check the draft
3. Improve, then repeat

THE MAP OF THE WHOLE COURSE

Five ways to make a model act like an investigator



Reflection

Draft, critique, revise.

LAB 1 · TODAY



Tool Use

Call real tools to inspect evidence.

LAB 2



ReAct

Reason a step, act, observe, repeat.

LAB 3



Planning

Break a hard question into steps.

LAB 4



Multiagent

Split roles, then reconcile findings.

LAB 5

The same idea in five shapes. You are starting with the simplest: the model reviewing itself.

QUALITY CONTROL, THE HUMAN WAY

How a report used to be checked



Analyst writes

A first report from the evidence.



Senior reviews

A second examiner marks overreach and gaps.



Analyst revises

The report is rewritten to match the evidence.



Sign-off

A defensible report, hours or days later.

CAREFUL, BUT IT DOES NOT SCALE

This loop works because a human reviewer reads every draft. That is exactly what breaks when a model can write a hundred reports in a minute.

THE PROBLEM

A confident answer is not a careful one

THE MODEL'S FIRST LINE

“The employee exfiltrated customers_q1.csv and delivered it to an outside party via Telegram and Gmail.”

WHY THAT SENTENCE IS A PROBLEM

No delivery is proven

The artifacts show sending activity, not that anyone received the file.

No recipient is confirmed

A handle is not a person, and there is no receipt.

A guess is stated as fact

“Exfiltrated” is a conclusion, written as if it were observed.

THE VOCABULARY THAT KEEPS YOU HONEST

Observation, inference, claim

Observation

What an artifact directly records.

IN THIS CASE

file_events.csv shows customers_q1.csv created at 01:05:22 UTC.

Inference

An interpretation that links observations.

IN THIS CASE

The file may have been prepared for sharing. Reasonable, but not recorded.

Claim

A checkable statement placed in the report.

IN THIS CASE

“Copied to Telegram at 01:07:41” is supported. “Delivered” is not.

SAY WHAT THE EVIDENCE SAYS

Cite it, or soften it

OVERREACHING		DEFENSIBLE
“was exfiltrated”	⇄	“the artifacts show it was copied and sent”
“was delivered to”	⇄	“this is consistent with an attempt to share”
“the suspect sent it”	⇄	“the device recorded an outbound message”
“proves theft”	⇄	“the current evidence does not confirm receipt”

The facts are identical. Only the certainty changes, and certainty is what a court tests.

THE GAP

Who checks the AI, in time?

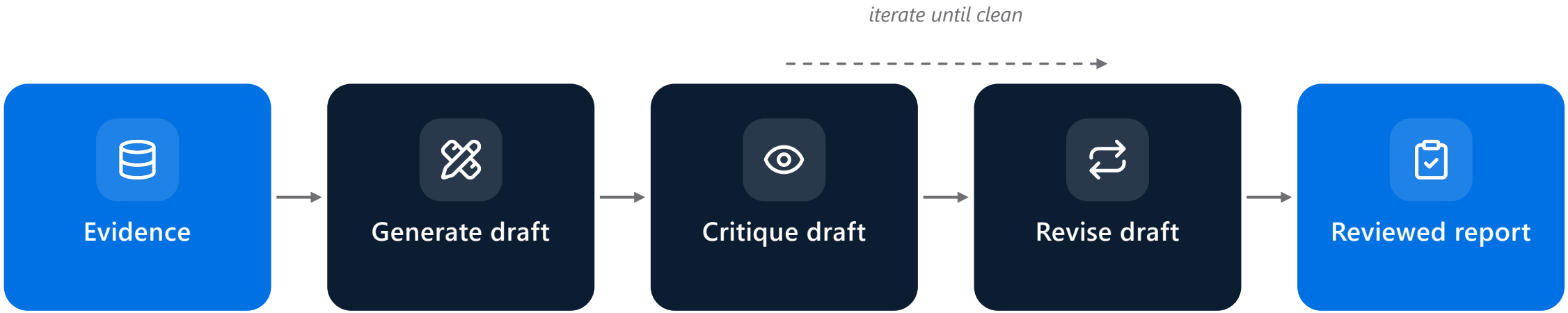


A MODEL CAN OUTRUN ITS REVIEWERS

One examiner can read a handful of drafts a day. A model can write hundreds in a minute. Peer review does not scale to that, and a single model call has no built-in check at all. The report ships with whatever overreach it started with.

THE ANSWER

The Reflection Pattern



WHEN DOES IT STOP?

The reviewer returns [[OK]] when nothing material is left to fix, or the loop hits its step limit.

WHAT YOU GET

A report whose claims have already survived one honest round of criticism before you ever read it.

THE TRICK BEHIND IT

One model, three jobs



The Analyst

Writes the report. Must cite a timestamp and an artifact for every claim.



The Reviewer

Reads only the draft. Flags overclaiming and weak uncertainty, or says [[OK]].



The Reviser

Takes the critique and rewrites, keeping the evidence in view.

IT IS THE SAME MODEL EACH TIME

Only the instructions change. The role it plays is set by the system prompt, which is why one model can both write and honestly criticize the same report.

THE INVESTIGATION

Case DF-2026-017

THE DEVICE

PHONE

Google Pixel 7, Android 14

ACQUIRED

Cellebrite UFED, full file system

WINDOW

overnight, 15 Feb 2026 (UTC)

THE FILE

customers_q1.csv

WHAT THE ARTIFACTS SHOW

A file appears

customers_q1.csv is created in Downloads and opened.

It moves and messages fly

The same file is copied into Telegram, then Telegram, Gmail, and network activity follow.

The original vanishes

customers_q1.csv is deleted from Downloads, all within about seven minutes.

THE EVIDENCE, MINUTE BY MINUTE

Nine events in about seven minutes

01:05:22	customers_q1.csv created in Downloads
01:06:12	same file accessed
01:07:41	same file hash copied into Telegram storage
01:07:54	outbound traffic to api.telegram.org
01:07:58	Telegram message: “sending that sheet now”
01:08:15	Telegram message: “delete after download”
01:09:39	outbound SMTP traffic via Gmail
01:09:44	Gmail message: “see attached export”
01:12:04	customers_q1.csv deleted from Downloads

Shaded rows are the ones people rush to over-read.

YOUR EVIDENCE PACKAGE

Six artifacts, one case

artifact_manifest

Case, device, and integrity metadata.

file_events

When files were created, copied, deleted, with hashes.

app_db_messages

Telegram, Gmail, and Slack message excerpts.

network_events

Outbound connections, bytes, and the app behind them.

location_events

Approximate device location over time.

chain_of_custody

Who handled the evidence, and hash verification.

SEE THE LOOP EARN ITS KEEP

One sentence, three passes

DRAFT v1

"The employee exfiltrated the file to an outside party through Telegram and Gmail."

CRITIQUE

Over-claimed. The artifacts support staging, outbound messaging, and network activity, not confirmed receipt. Separate observed events from inferred transmission.

REVISED v2

"Artifacts show the file was copied into Telegram storage, followed by outbound Telegram and Gmail activity consistent with attempted sharing. The evidence does not confirm delivery to a recipient."

KEEP THIS STRAIGHT

The agent drafts. You decide.

It speeds the self-check

The obvious overclaims are gone before a human reads the report.

It flags, it does not rule

The critique points at risk. It does not establish guilt or innocence.

It never signs off

A qualified examiner owns the final forensic judgment, always.

THE HONEST FRAMING

Reflection makes a model a better assistant to an examiner. It does not make the model the examiner.

PRACTICAL

Your turn, in the browser

1	**Open in Colab**	Click the badge in the notebook. It opens in your browser, nothing to install.
2	**Paste a free key**	Get a free Groq key, paste it when asked. It is hidden and not saved.
3	**Run Part A by hand**	Generate the draft, read the critique, see the revision.
4	**Run Part B**	Let ReflectionAgent loop the whole thing until it is clean.
5	**Answer four questions**	Trace a claim to its artifact, and name what stays uncertain.

SectraTEK

Reflection is the first habit of a careful AI.

Next lab: Tool Use, where the agent stops guessing and starts inspecting the artifacts itself.

